

AI-POWERED VIRTUAL TEACHER FOR PRIMARY EDUCATION

**Based on Sustainability Goal 4 :
Quality Education**

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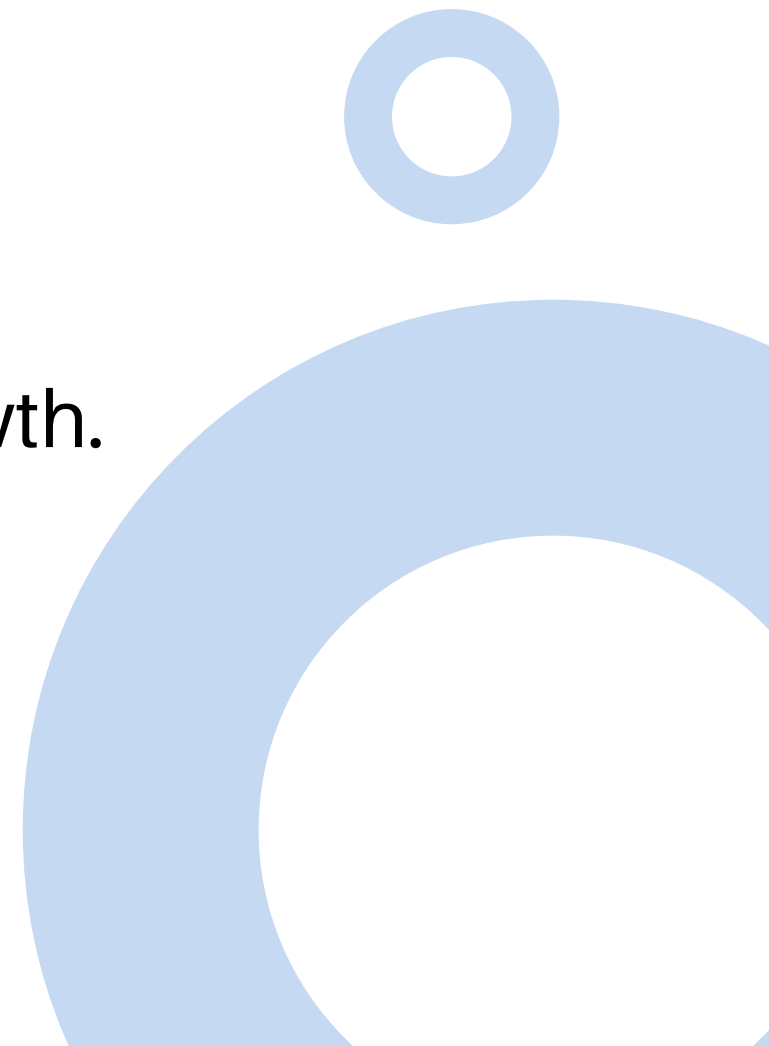
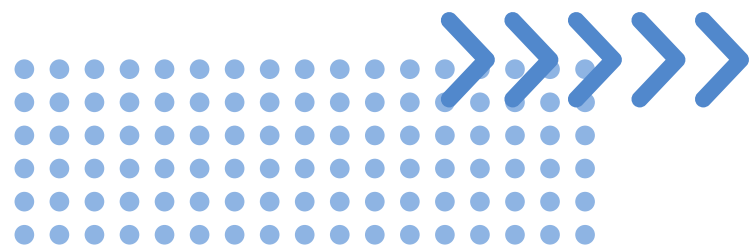
Introduction

What is the Sustainability Goal 4 : Quality Education

“Ensure inclusive and equitable education and promote lifelong learning opportunities for all”

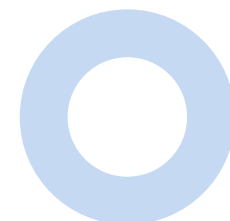
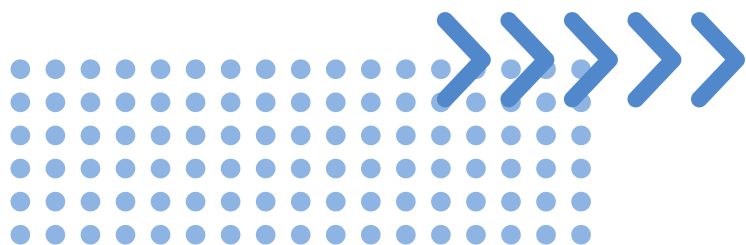
Problem Statement

- Millions of children in India lack access to quality primary education.
- Rural & underserved regions face teacher shortages.
- Foundational learning gaps in literacy and numeracy affect long-term growth.
- Existing e-learning solutions lack personalization and interactive teaching.



Objectives

- Build an AI-powered virtual teacher for Grades 1–3.
- Provide curriculum-aligned lessons and doubt clarification.
- Support reference material uploads for context-specific learning.
- Enable voice-based interaction for natural teacher-student dialogue.
- Use teacher avatars (male & female) for engaging delivery.
- Align with UN Sustainable Development Goal 4 – Quality Education.



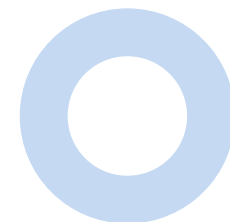
Scope

Target Audience :

- Indian Primary school students (Grades 1–3) studying in English medium.
- Special emphasis on children without access to physical schooling.

Features in Scope:

- AI-based virtual teacher powered by Groq Llama model (llama-3.3-70b-versatile) with RAG (Retrieval Augmented Generation).
- Curriculum-aligned knowledge base (vector databases created from NCERT/other textbooks).
- Students can upload PDFs or learning materials for contextual doubt-solving.
- Voice input/output for natural interaction.
- Teacher avatars (male/female) to simulate classroom-like teaching.



Progress till Interim Review

Vector database

Created a pipeline (vectordb.py) to convert locally stored textbooks into vector embeddings using ChromaDB and sentence-transformers.

Implemented chunking and metadata storage (grade, subject, chapter).

AI Chatbot Engine:

Built a FastAPI backend integrated with Groq API (Llama-3.3-70B).

Uses RAG to fetch relevant content from vector databases and uploaded user material.

Guardrails: If no relevant context is found, AI politely declines instead of hallucinating.

Supports doubt-solving with contextual accuracy.

Frontend (React + CSS):

Developed a functional chatbot interface where students can:

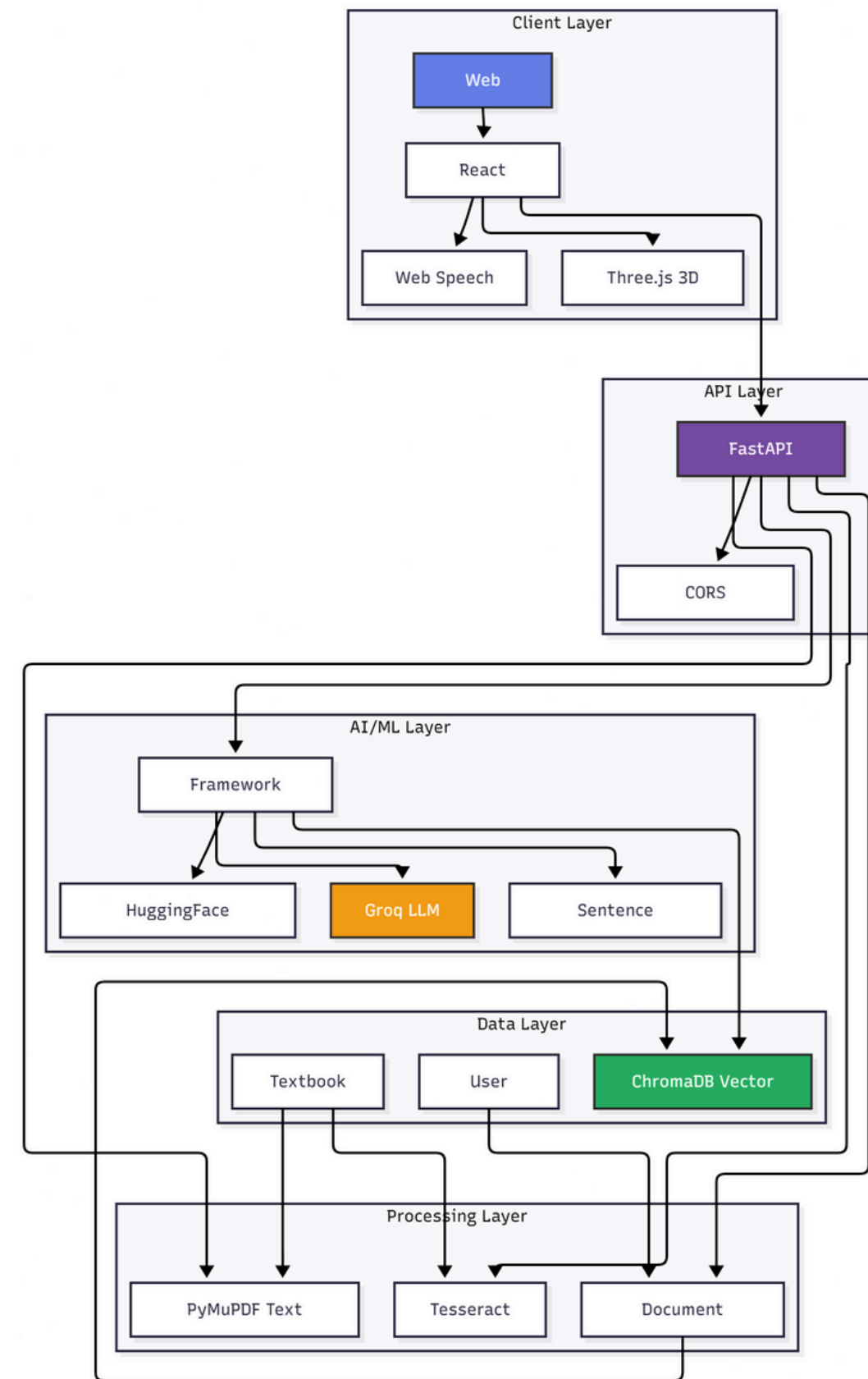
- Select grade (1-3).
- Choose subject (Maths, English, EVS).
- Upload study materials (optional).
- Ask questions and receive AI teacher responses in a clean chat format.

The UI design is child-friendly, with gradients, rounded components, and responsive cards.

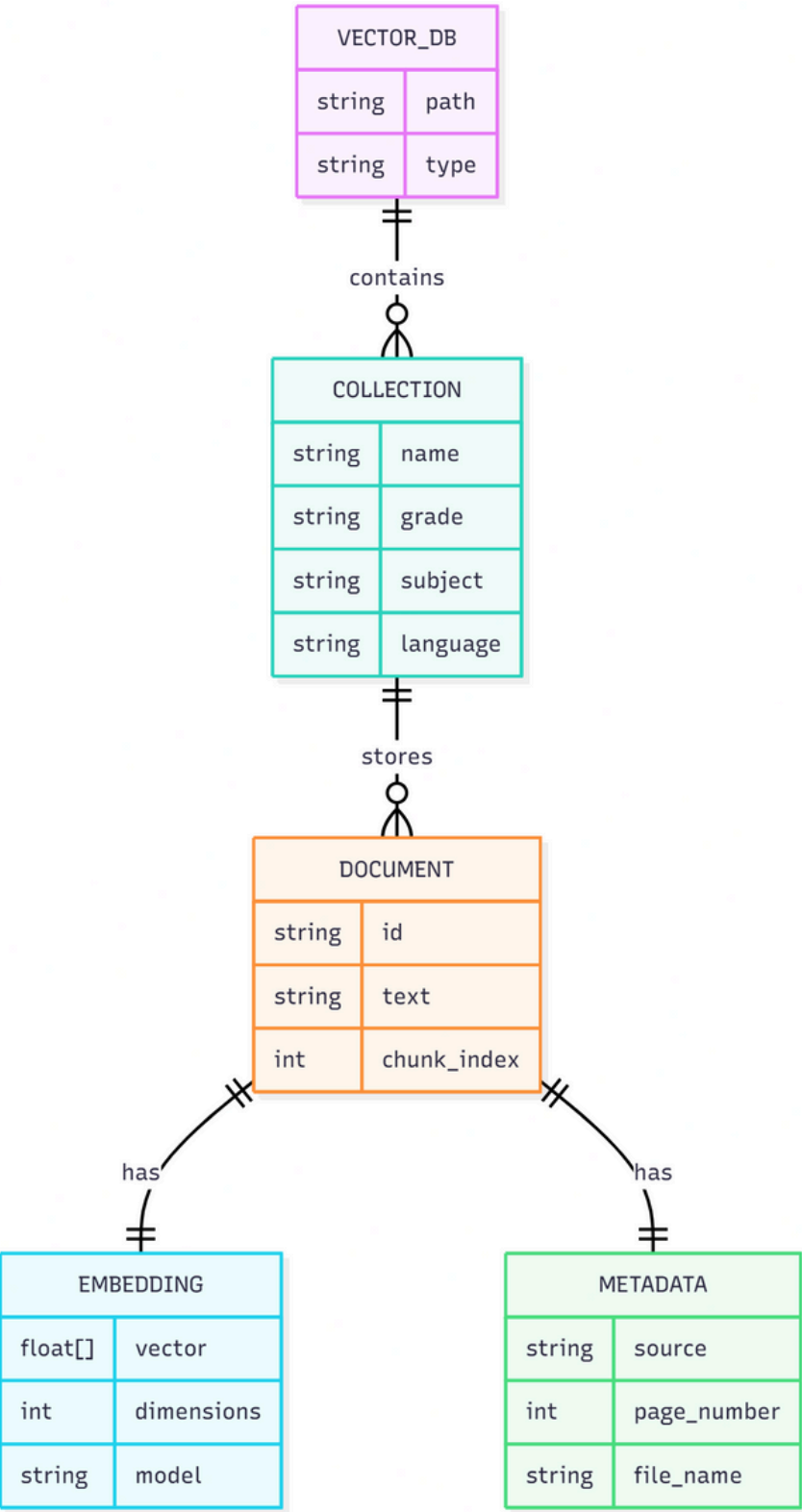
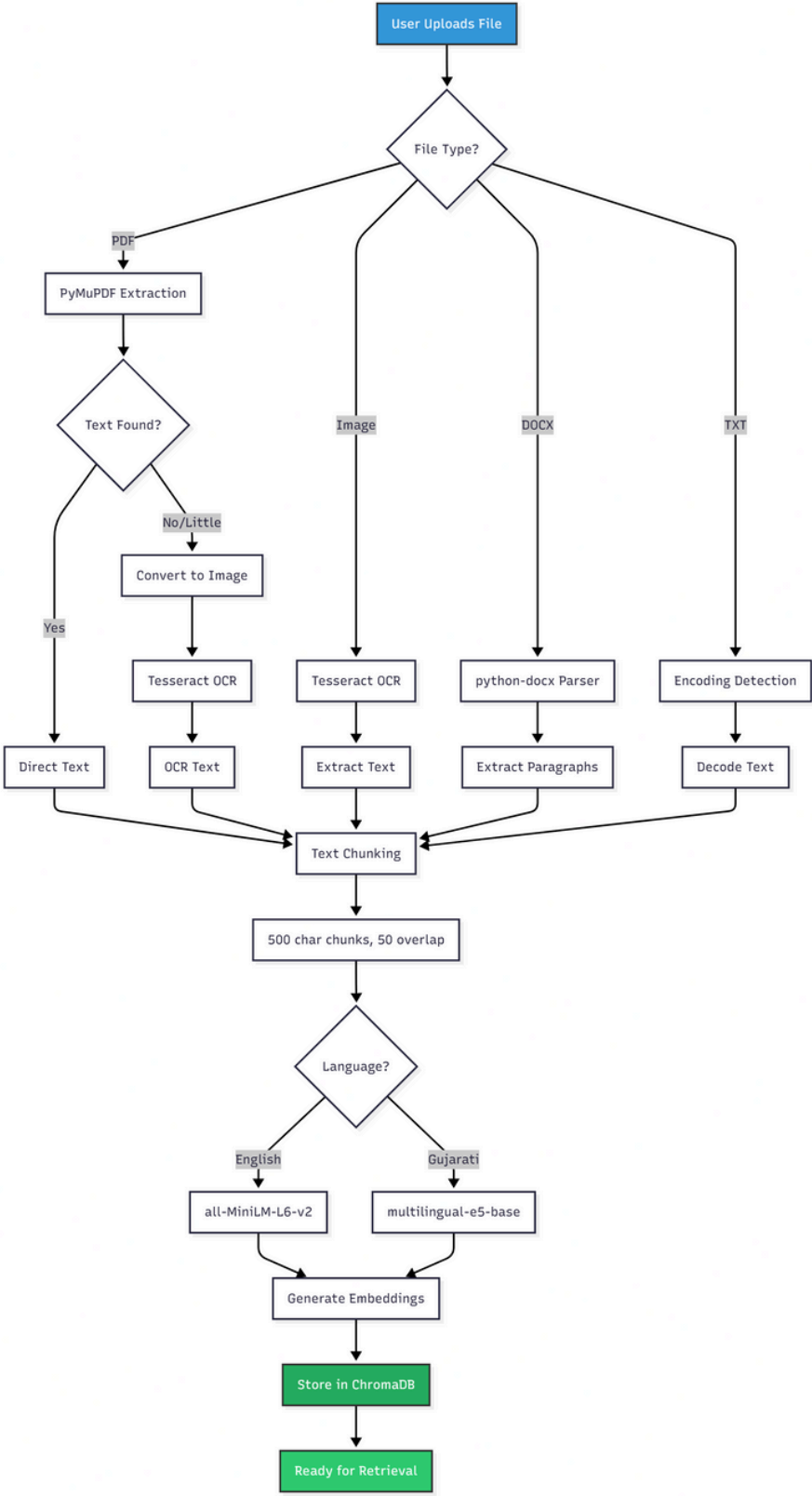
Implementation

- Voice Integration: Added microphone input so students can ask doubts via speech.
- Multi-Language communication : The student can communicate in 3 languages - English, Hindi, Gujarati
- Integrated text-to-speech (TTS) for AI teacher responses.
- Teacher Avatars: Introduced animated male and female avatars for immersive teaching.
- Gestures : The avatars interact using different gestures for a more realistic experience.
- Curriculum Expansion: Added Gujarati Curriculum from grade 1 to 3.

System Architecture

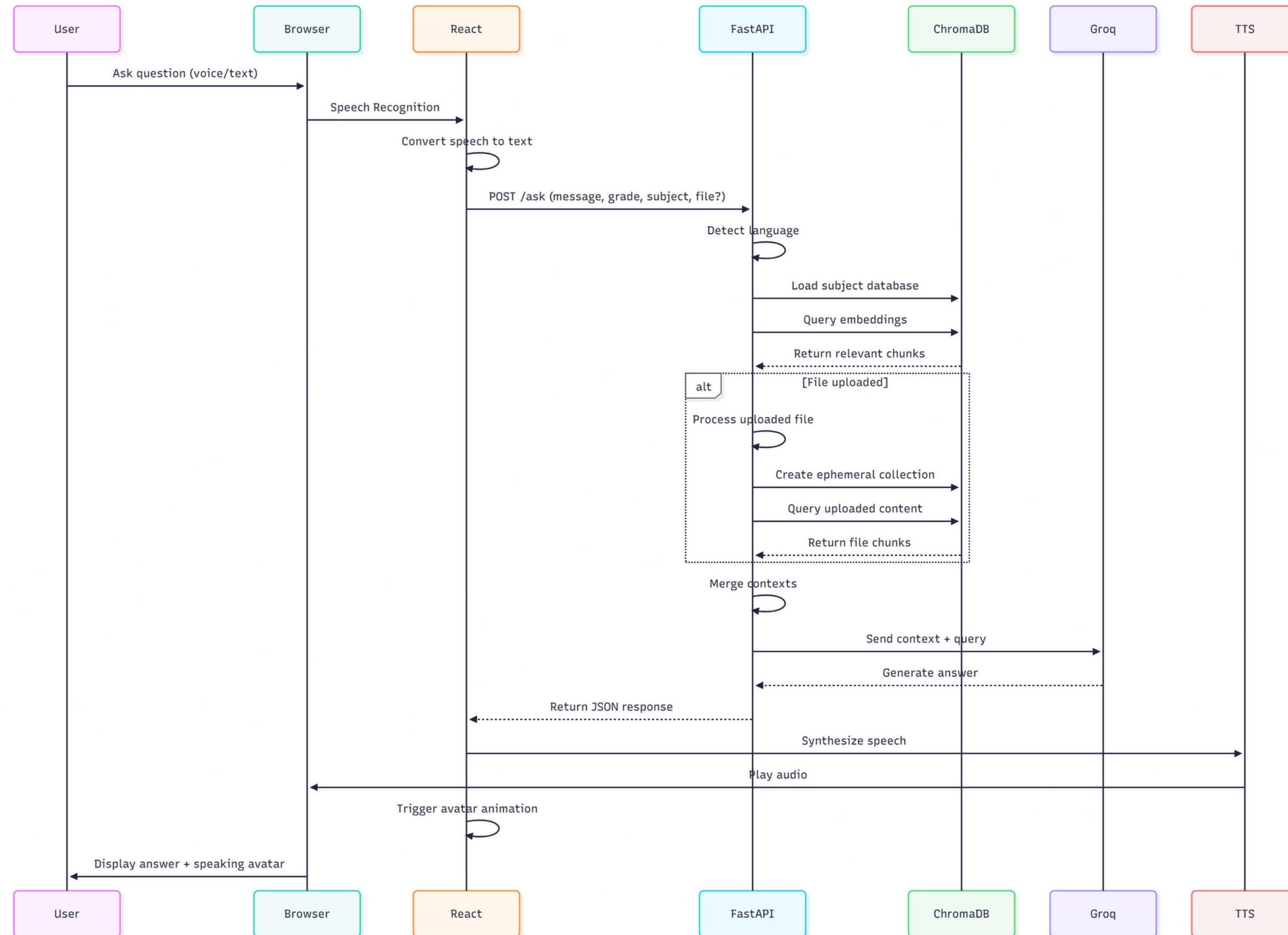


Database Architecture

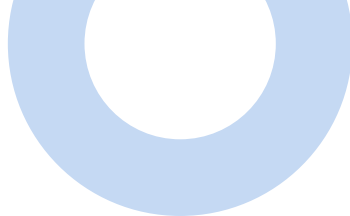


Document Processing

Userflow



Dependencies



Service	Purpose	Version/Model	Cost
Groq API	LLM Inference	llama-3.3-70b-versatile	Free tier available
HuggingFace	Embedding Models	sentence-transformers/all-MiniLM-L6-v2	Free (open-source)
HuggingFace	Multilingual Embeddings	intfloat/multilingual-e5-base	Free (open-source)
Tesseract OCR	Text Extraction	5.x with Gujarati support	Free (open-source)
Web Speech API	Voice I/O	Browser native	Free (browser built-in)

Internal Dependencies



1 Frontend Dependencies

React Three Fiber (@react-three/fiber 9.4.0)

- Purpose: React renderer for Three.js
- Key Features Used:
 - Canvas component for WebGL context
 - useFrame hook for animation loop
 - Declarative 3D scene composition

React Three Drei (@react-three/drei 10.7.6)

- Purpose: Helper utilities for React Three Fiber
- Key Features Used:
 - useGLTF - GLB model loader
 - useFBX - FBX animation loader
 - useAnimations - For using animations
 - OrbitControls - Camera controls
 - PerspectiveCamera - Camera setup / Lighting

Three.js (0.180.0)

- Purpose: Core 3D graphics library
- Key Features Used:
 - SkeletonUtils - Skeleton cloning for instancing
 - KeyframeTrack - Animation track creation
 - LoopRepeat - Animation looping
 - SkinnedMesh - Rigged character rendering
 - Lighting system (ambient, directional, spot, point)

2 Backend Dependencies

FastAPI (0.115.8)

- Purpose: Modern Python web framework
- Key Features Used:
 - Async request handling
 - Form data parsing (Form, UploadFile)
 - Automatic API documentation
 - CORS middleware

LangChain Components Used:

- HuggingFaceEmbeddings - Wrapper for sentence-transformers
- Chroma - Vector store wrapper with persistence
- similarity_search() - Semantic search method
- Document chunking utilities

Sentence Transformers (5.1.0)

- Purpose: State-of-the-art sentence embeddings
- Models Used:
 - sentence-transformers/all-MiniLM-L6-v2 (English)
 - Dimensions: 384
 - Speed: Fast
 - Use case: English textbook embeddings
 - intfloat/multilingual-e5-base (Multilingual)
 - Dimensions: 768
 - Languages: 100+ including Gujarati
 - Use case: Gujarati textbook embeddings

Internal Dependencies

Document Processing Libraries

PyMuPDF (1.25.2)

- Purpose: PDF text extraction
- Key Features Used:
 - `fitz.open()` - PDF document loading
 - `page.get_text()` - Direct text extraction
 - `page.get_pixmap()` - Page to image conversion

Pytesseract (0.3.13)

- Purpose: Python wrapper for Tesseract OCR
- Key Features Used:
 - `image_to_string()` - OCR text extraction
 - Multi-language support (`lang='guj+eng'`)
 - Image preprocessing integration

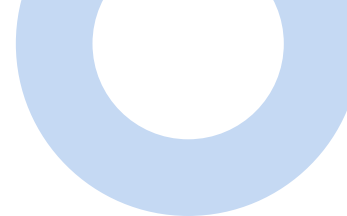
Pillow (10.4.0)

- Purpose: Image processing
- Key Features Used:
 - Format conversion (PNG, JPEG, etc.)
 - Purpose: DOCX file parsing
 - Key Features Used:
 - `Document()` - DOCX loading
 - Paragraph extraction
 - Text formatting preservation

ChromaDB (1.0.20)

- Purpose: Open-source vector database
- Key Features Used:
 - `PersistentClient` - Disk-based storage
 - `EphemeralClient` - In-memory temporary storage
 - `get_or_create_collection()` - Collection management
 - `query()` - Vector similarity search
 - `add()` - Document insertion with embeddings





External Assets

1. 3D Models (GLB Format)

- Source: Ready Player Me (RPM) or custom creation
- Format: GLB (GL Transmission Format Binary)

Files:

- teacher_female.glb - Female teacher avatar
- teacher_male.glb - Male teacher avatar
- Components:
 - Skinned meshes (Wolf3D_Hair, Wolf3D_Body, Wolf3D_Outfit_Top, etc.)
 - Skeleton (Hips root bone)
 - Morph targets (facial expressions)
 - Materials (PBR textures)

2. Animations (FBX Format)

- Source: Mixamo (Adobe's free animation library)
- Format: FBX (Filmbox)

Files:

- Idle_Standing_female.fbx - Female idle animation
- Talking_Standing_female.fbx - Female talking animation
- Idle_Standing_male.fbx - Male idle animation
- Talking_Standing_male.fbx - Male talking animation
- Skeleton: Mixamorig (requires retargeting to RPM skeleton)
- Animation Properties:
 - Loop: Repeat
 - Transition: 0.5s fade in/out
 - Tracks: Position, rotation, scale keyframes

Choose your AI Teacher

Select your preferred teacher avatar



Ms. Shanti
Female Teacher



Mr. Raj
Male Teacher

Avatar Selection

Chat with your AI Teacher / તમારા એઆઈ શિક્ષક સાથે ચેટ કરો

Hello! I'm your AI teacher. Select your grade and subject above, then ask me anything!

Type your question here or use voice input...

Send

AI Teacher Bot | એઆઈ શિક્ષક બોટ

Select Your Grade / તમારો ધોરણ પસંદ કરો

Grade 3 / ધોરણ 3

Choose Medium / માધ્યમ પસંદ કરો

☐ English Medium

☒ ગુજરાતી માધ્યમ (Gujarati Medium)

Choose Subject / વિષય પસંદ કરો

☒ English

☒ Maths

☐ Environmental Studies

Voice Language / અવાજની ભાષા

☐ English

☒ હિન્દી (Hindi)

☒ ગુજરાતી (Gujarati)

Upload Learning Materials (Optional) / શીખવાની સામગ્રી અપલોડ કરો

 [Click to upload documents, images, or videos](#)

 Chat with your AI Teacher / તમારા એઆઈ શિક્ષક સાથે ચેટ કરો

 Hello! I'm your AI teacher. Select your grade and subject above, then ask me anything!



Type your question here or use voice input...

Send

Working Model

AI Teacher Bot

Select Your Grade

Grade 3

Choose Subject

● English

● Maths

○ Environmental Studies

Upload Learning Materials (Optional)

📎 SpaceRangers-solar-system.pdf

💬 Chat with your AI Teacher

Example

- Grade 3 student
- Subject Chosen - EVS
- asked questions from reference material and textbook



💬 Chat with your AI Teacher / તમારા એઆઈ શિક્ષક સાથે ચેટ કરો

which are the herbs mentioned in the chapter "getting to know plants"

Let's learn about herbs. In the chapter "Getting to Know Plants", we came across some interesting information about herbs. Herbs are small plants with soft stems that don't become hard or woody. We found out that some examples of herbs are: 1. Mint - This is a common herb that many people have in their homes. It's often used in drinks and food. 2. Tomato - Yes, you read that right! Tomato is also a type of herb because it has a soft stem. 3. Coriander - This is another herb that people use in cooking. 4. Mustard - We use mustard seeds in our food, and the plant itself is also an herb. We also learned that grasses are a type of herb. They have thin, flat leaves and hollow stems. There are many different kinds of grasses, and you can find them all around you. So, these are the herbs mentioned in the chapter "Getting to Know Plants". I hope this helps you learn more about the wonderful world of plants!



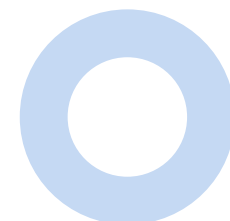
Type your question here or use voice input...

Send



Future Directions

- Sync voice + lip movement for realistic classroom simulation.
- Classroom environment background to make it look like a real classroom instead of an artificial teacher chatbot.
- Include more languages and curriculums so that every student in India can access to quality education.



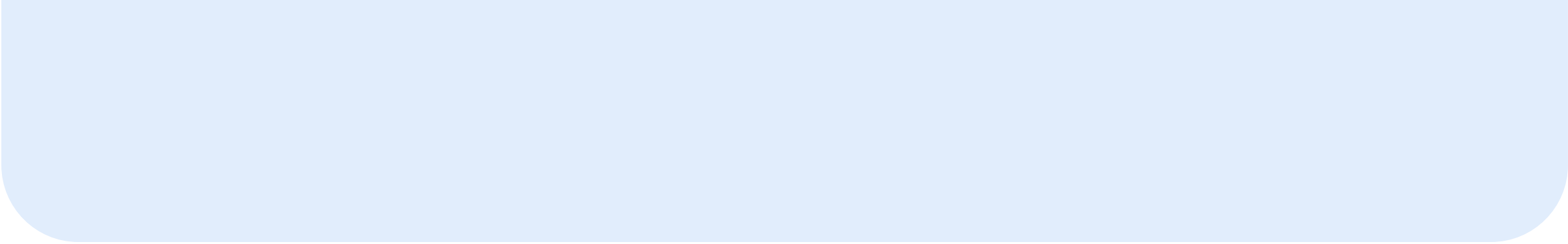


Conclusion

This project is more than a doubt-solving chatbot, it is a virtual teacher for children deprived of schooling opportunities. By combining AI, RAG, Groq LLM, and interactive avatars, the platform aspires to create a personalized, scalable, and accessible solution for foundational learning in India.

The vision aligns with United Nations Sustainable Development Goal 4 (Quality Education), aiming to ensure inclusive and equitable education for all, particularly those who need it the most.





**THANK
YOU!**

